

Population Growth and Economic Development: Causes, Consequences, and Controversies

Chapter 6

Econ 322 – The Global Economy: Trade and Development

University of Tennessee, Knoxville

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Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

6.1 Population Growth: Past, Present, and Future

- ▶ Structure of the world's population
 - ▶ Population growth in history
 - ▶ Geographic region
 - ▶ Fertility and mortality trends
 - ▶ Rate of population increase
 - ▶ Birth rates, death rates, total fertility rates
 - ▶ Age structure and dependency burdens

Birth rate: births per 1,000 people. **Death rate:** deaths per 1,000 people. **Total fertility rate (TFR):** average number of children per woman; a TFR of about 2.1 is the replacement level.



Street crowds in Dhaka, Bangladesh, one of the world's most densely populated cities.

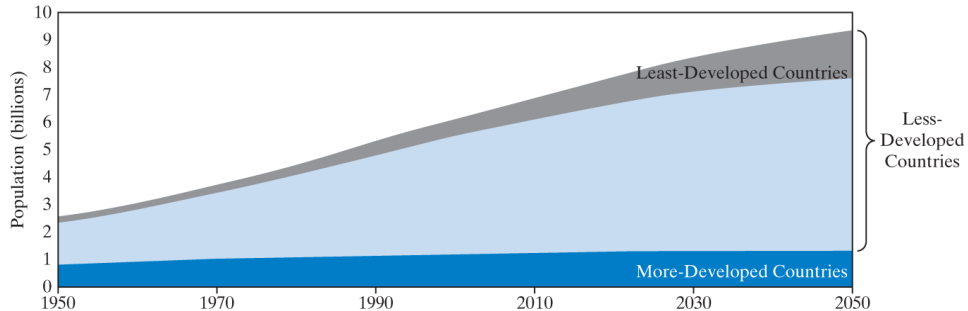
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Table 6.1 Estimated World Population Growth

Year	Population (millions)	Annual Increase (%)
8000 BCE	5	
2000 BCE	25	0.03
1000 BCE	50	0.07
1 CE	175	0.13
1000	250	0.04
1500	450	0.12
1800	950	0.25
1850	1 250	0.55
1900	1 650	0.56
1950	2 536	0.86
1960	3 033	1.81
1970	3 700	2.01
1980	4 458	1.88
1990	5 331	1.8
2000	6 145	1.43
2010	6 958	1.25
2018	7 621	1.14
2050 (forecast)	9 800	0.79
2100 (forecast)	11 200	0.27

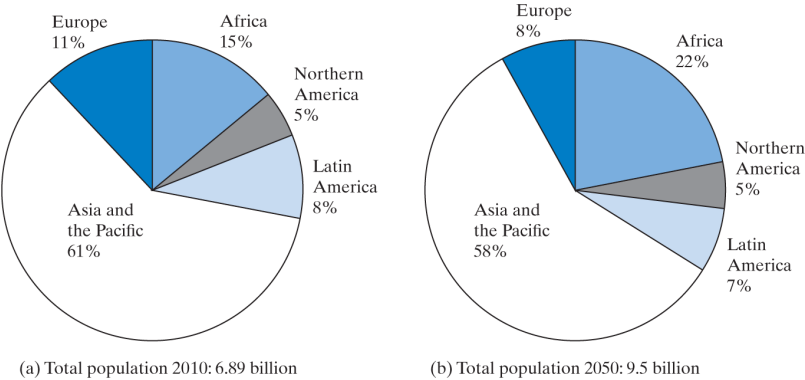
Sources: Selected historical estimates based on data from the United Nations Population Division, the US Census Bureau, and the Population Reference Bureau. Data starting in 1950 are from the UN Population Bureau.

Figure 6.1 World Population Growth, 1950–2050



Source: Population Reference Bureau World Population Data Sheet 2012, page 4; data are drawn from United Nations Population Division, World Population Prospects: The 2010 Revision (2011), medium-variant estimates.

Figure 6.2 World Population by Region, 2010 and 2050: Africa's Growing Share



Source: Data from Population Reference Bureau, World Population DataSheet, 2010.

Figure 6.3 Map with Country Sizes Proportional to Share of World Population



Source: [worldmapper.org:http://www.worldmapper.org/display.php?selected=2](http://www.worldmapper.org/display.php?selected=2)

Table 6.2 Births Per Woman: Fertility Rate for Selected Countries, 1990 and 2017 (1/2)

	1990	2017
Fertility High and Falling Relatively Slowly		
Niger	7.8	7.2
Somalia	7.4	6.2
Congo (DRC)	6.7	6
Mali	7.2	6
Chad	7.3	5.8
Angola	7.2	5.6
Burundi	7.5	5.6
Fertility Falling Relatively Rapidly		
Afghanistan	7.5	4.5
Ethiopia	7.2	4.1
Rwanda	7.2	3.8
Pakistan	6	3.4
Lao PDR	6.2	2.6
Saudi Arabia	5.9	2.5

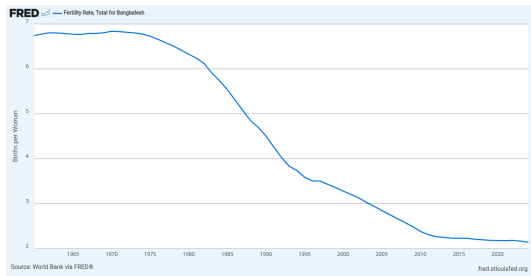
Source: *World Development Indicators, Table 2.14, 2018, accessed 23 May 2019. Abbreviations: LMC—Lower-middle-income country; UMC—Upper-middle-income country.*

Table 6.2 Births Per Woman: Fertility Rate for Selected Countries, 1990 and 2017 (2/2)

	1990	2017
Close to Replacement Fertility		
Bangladesh	4.5	2.1
Nepal	5.2	2.1
Turkey	3.1	2
Jamaica	2.9	2
Malaysia	3.6	2
Sri Lanka	2.5	2
Below Replacement Fertility		
China	2.4	1.6
Japan	1.5	1.4
Poland	2.1	1.4
Portugal	1.6	1.4
Moldova	2.4	1.2
South Korea	1.6	1.1
Country Group Averages		
World	3.3	2.4
Low income	6.3	4.6
LMC	4.2	2.7
UMC	2.6	1.8
High income	1.9	1.7

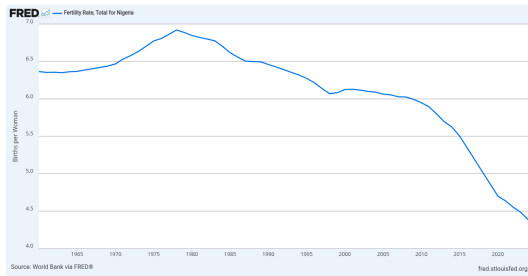
Source: World Development Indicators, Table 2.14, 2018, accessed 23 May 2019. Abbreviations:
LMC—Lower-middle-income country; UMC—Upper-middle-income country.

Two Fertility Paths: Bangladesh and Nigeria, 1960–2023



Bangladesh: births per woman fell from almost 7 to about 2, close to replacement (Table 6.2).

Source: FRED, Federal Reserve Bank of St. Louis



Nigeria: fertility stayed above 6 for decades and is still above 4, so its population keeps growing rapidly.

Source: FRED, Federal Reserve Bank of St. Louis

6.2 Population Growth: Past, Present, and Future

- ▶ The Hidden Momentum of Population Growth
 - ▶ High birth rates cannot be altered overnight
 - ▶ Age structure of developing country populations

Demographic momentum: even if every woman suddenly had only two children, the population would keep growing for decades, because a wide-based age pyramid means today's many children will themselves become parents. Planning for schools, housing and jobs must look at the age structure, not just at current fertility.

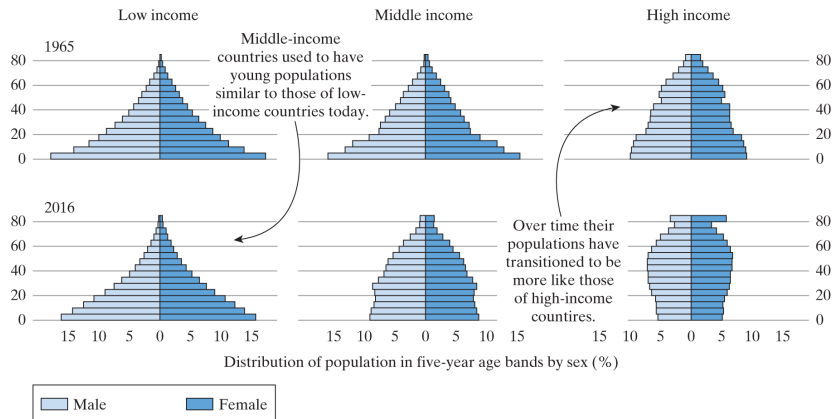
Some findings in the UN World Population Prospects 2017 Revision

- ▶ World population is now projected to reach 8.6 billion by 2030; 9.8 billion by 2050; and 11.2 billion by 2100. (Projected population sizes have actually been increasing in recent reports, even as population growth has slowed.)
- ▶ Most population growth will continue to occur in developing regions
- ▶ “Give or take a billion” - Projections depend upon assumptions
- ▶ Even at 80% confidence:
 - ▶ 2050 world population could be as few as 9.5 billion or as many as 10 billion
 - ▶ 2100 world population could be as low as 10.0 billion to as high as 12.4 billion
- ▶ India will become the most populous country, passing China sooner than previously projected - in about 2024, each will have about 1.44 billion people
- ▶ Nigeria's population is projected to pass the United States shortly before 2050; by 2100 it could pass China as the second most populous country (after India)

Some findings in the UN World Population Prospects 2017 Revision (Continued)

- ▶ Population growth will be concentrated in **Africa**
 - ▶ From 2017-2050, populations of 26 African countries are projected to at least double
- ▶ The 47 least developed countries are **projected to double** from about **one billion in 2013 to 1.9 billion in 2050**
 - ▶ These 47 countries had fertility rates of 4.3 births per woman in 2010-2015
- ▶ Projected total population is higher in this and the last major reports, particularly after 2075, because:
 - ▶ Fertility estimates have been estimated to be higher in some countries, especially in high-fertility sub-Saharan African countries, increasing estimated births per woman
 - ▶ In some cases, the actual level of fertility appears to have risen in recent years; in some other cases, the previous estimate was too low
- ▶ In contrast, during 2010-2015, fertility was below replacement level in 83 countries, comprising about 46% of the global population

Figure 6.4 Population Pyramids by Income Group, 1965 and 2016



Source: 2018 Atlas of SDGs (World Bank, 2018), section 3, p. 10. Available at:

<http://documents.worldbank.org/curated/>

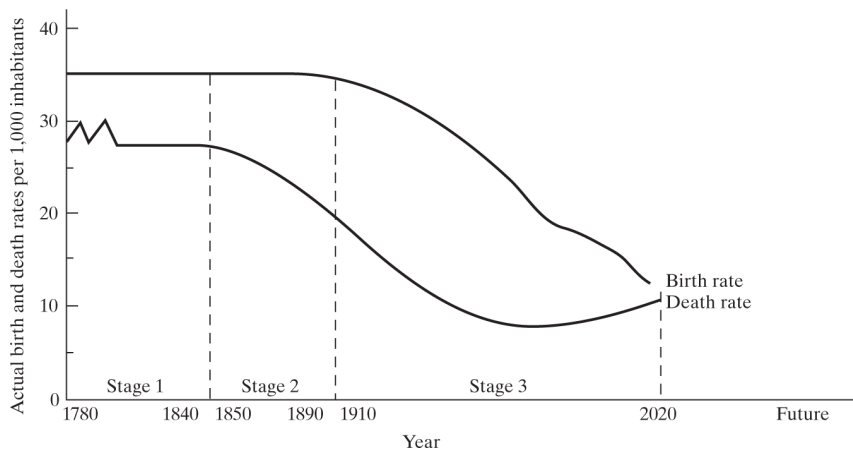
<en/590681527864542864/Atlas-of-Sustainable-Development-Goals-2018-World-Development-Indicators>

6.3 The Demographic Transition

- ▶ Stage I: High birthrates and death rates
- ▶ Stage II: Continued high birthrates, declining death rates
- ▶ Stage III: Falling birthrates and death rates, eventually stabilizing

Rapid population growth happens in Stage II: death rates fall with better health care, nutrition and sanitation, while families keep having the same number of children. Many developing countries are in this stage today; birthrates fall in Stage III as education (especially for women), contraception and changing economic values spread.

Figure 6.5 The Demographic Transition in Western Europe



Birth and death rates per 1,000 in Western Europe: the gap between the two curves in Stage II is the period of rapid natural increase.

Source: *Todaro and Smith (2020), Figure 6.5*

6.4 The Causes of High Fertility: The Malthusian Model

- ▶ The Malthusian Population Trap
 - ▶ The idea that rising population and diminishing returns to fixed factors result in low levels of living (population trap)

Population tends to grow faster than the food supply; more people on fixed land means diminishing returns, pushing income per person back down until it hits the subsistence level: the population trap.

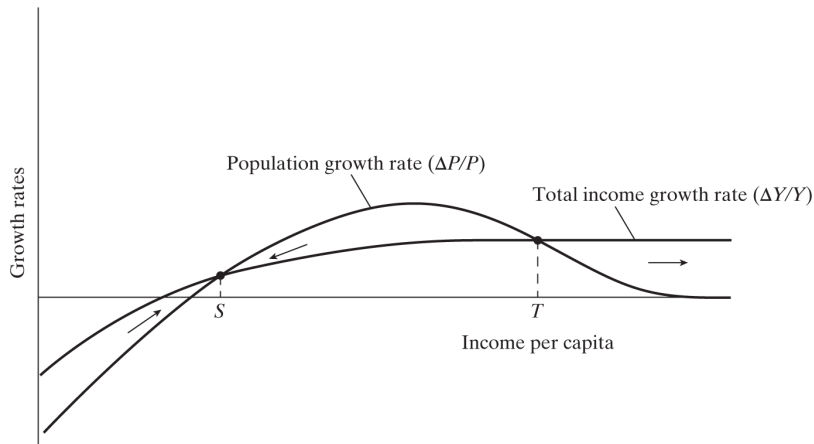


Thomas Robert Malthus (1766–1834), author of the *Essay on the Principle of Population* (1798).

Wikimedia Commons, John Linnell, CC

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Figure 6.6 The Malthusian Population Trap



Population growth and total income growth against income per capita: the economy is pulled back to the low-level equilibrium wherever population grows faster than income.

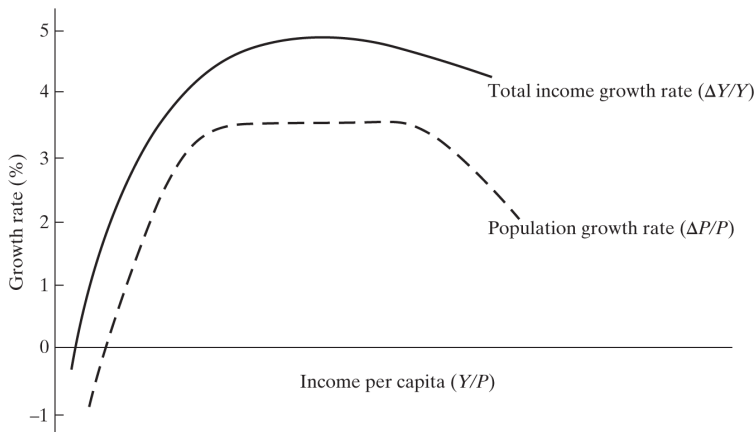
Source: *Todaro and Smith (2020), Figure 6.6*

Criticisms of the Malthusian Model

- ▶ Overlooks impact of technological progress
- ▶ Currently no positive correlation between population growth and levels of per capita income in the data
- ▶ Overlooks microeconomics of family size; importance of examining individual more than aggregate variables

Why the trap has been escaped: technological progress raises productivity faster than population grows (Figure 6.7), and better education and health care reduce both mortality and fertility. In rural or fragile regions with poor infrastructure and governance, Malthusian-like conditions may still appear.

Figure 6.7 How Technological and Social Progress Avoids the Population Trap



Technological progress shifts the income growth curve up, and social progress lowers the population growth curve, so income per capita can keep rising.

Source: *Todaro and Smith (2020), Figure 6.7*

Why We Continue to Study the Malthusian Trap

- ▶ First, many believe that the Malthusian trap holds in poor countries today, despite recent evidence
 - ▶ Those engaged in economic development should understand the model – and elements of it that do not currently apply – so they can engage the debate effectively
- ▶ Second, it seems clear that such traps have occurred in (pre-)historical past; possible factors in population collapses, including in the pre-Columbian Americas
- ▶ Third, reasons this model no longer applies underline the importance of factors that can prevent its (re-) emergence. These include:
- ▶ Efforts to sustainably raise agricultural productivity
- ▶ Increasing women's empowerment and freedom to choose—along with increased incomes—which reduce the old-age security motive behind high fertility
- ▶ We examine population policy analysis in more detail shortly...

6.4 The Microeconomic Household Theory of Fertility

- ▶ The Microeconomic Household Theory of Fertility
- ▶ The Demand for Children in Developing Countries
 - ▶ First two or three as “consumer goods”
 - ▶ Additional children as “investment goods”:
 - ▶ Work on family farm, microenterprise
 - ▶ Old age security motivation

The Demand for Children Equation

Demand for Children Equation

$$C_d = f(Y, P_c, P_x, t_x), \quad x = 1, \dots, n$$

where

- ▶ C_d is the demand for surviving children
- ▶ Y is the level of household income
- ▶ P_c is the “net” price of children
- ▶ P_x is the price of all other goods
- ▶ t_x is the “tastes” for goods relative to children

The Demand for Children Equation (Continued)

“Demand for Children” Equation, Continued

$$C_d = f(Y, P_c, P_x, t_x), \quad x = 1, \dots, n$$

Under neoclassical conditions, we would expect:

- ▶ The higher the household income, the greater the demand for children
- ▶ The higher the net price of children, the lower the quantity demanded
- ▶ The higher the prices of all other goods relative to children, the greater the quantity of children demanded
- ▶ The greater the strength of tastes for goods relative to children, the fewer children demanded

Note: In symbols these relationships may be written as:

$$\frac{\partial C_d}{\partial Y} > 0, \quad \frac{\partial C_d}{\partial P_c} < 0, \quad \frac{\partial C_d}{\partial P_x} > 0, \quad \frac{\partial C_d}{\partial t_x} < 0$$

Microeconomic Theory of Fertility: Some Implications

- ▶ It follows logically that fertility is lower where there is:
 - ▶ More female nonagricultural employment, at higher wages
 - ▶ Raised women's education, role, and status
 - ▶ Expanded schooling opportunities, with lowered real costs
 - ▶ Increases in family income levels
 - ▶ Reduction in infant mortality, better health care
 - ▶ Development of old-age and social security plans
- ▶ There is empirical evidence of varying strengths to support these conclusions



A classroom in Ghana: girls' schooling raises the opportunity cost of children and is strongly linked to lower fertility.

Wikimedia Commons, Amuzujoe, CC BY-SA 4.0

Microeconomic Theory of Fertility: Some Implications (Continued)

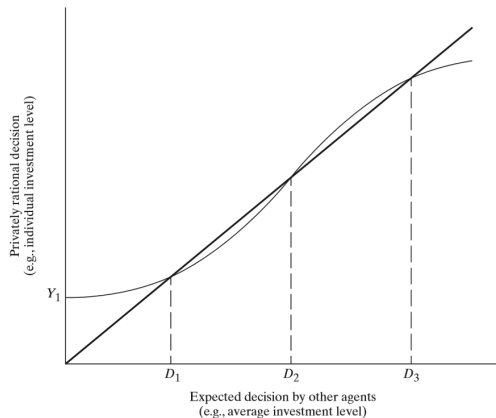
- ▶ Additional influences could be important in affecting fertility, e.g.:
 - ▶ Lowered prices and better information on contraceptives
 - ▶ Policies that have the effect of reducing boy preference
 - ▶ Direct incentives such as subsidy benefits
 - ▶ Note: This starts to give shape to what could be effective as *population policies*
- ▶ In addition, social norms and fertility expectations can play significant roles in different contexts

Rationale for Population Policy: Are There Market Failures?

- ▶ The standard question: Are there market failures...
- ▶ *Coordination failures ("Figure 4.1 Reinterpreted")*
- ▶ Environmental impacts
- ▶ Imperfect information (including information lags)
- ▶ Financial market failures
- ▶ Note: Sociological factors also play a role via social norms and status; example: external costs imposed by family status seeking through activities of many sons

Reinterpreting Figure 4.1 Multiple Equilibria in Fertility Choice

- ▶ Now on x-axis: expected number of children in other families
- ▶ Now on y-axis: a family's best response, in its own fertility choice, to the expected number of children of other families



Multiple equilibria: where the S-shaped best-response curve crosses the 45-degree line.

Source: *Todaro and Smith (2020), Figure 4.1, reinterpreted*

Figure 4.1 Reinterpreted: Why the Best-Response Curve Is S-Shaped

- ▶ General upward slope (showing *complementarity*) may reflect the impact of social norms about fertility
- ▶ The curve increasing at an *increasing* rate may be due to:
 - ▶ Response to negative impact of average fertility on wages, and/or
 - ▶ Response to negative impact on probability any one child will get modern sector employment
 - ▶ Presence of older children who can take care of younger siblings
- ▶ The curve subsequently increasing at a *decreasing* rate may be due to:
 - ▶ Impact on supply of education and health goods
 - ▶ Costs of raising more children rising relative to benefits

The Debate on High Fertility, 1: “It’s Not a Real Problem”

- ▶ “The real problem is not population growth but the following”:
 - ▶ Underdevelopment
 - ▶ World resource depletion and environmental destruction
 - ▶ Population distribution
 - ▶ Subordination of women
- ▶ “Overpopulation is a Deliberately Contrived False Issue”
- ▶ “Population Growth is a Desirable Phenomenon”

The Debate on High Fertility, 2: “Population Growth Is a Real Problem”

- ▶ Extremist arguments
- ▶ Theoretical arguments
- ▶ Empirical arguments:
 - ▶ Lower economic growth
 - ▶ Poverty
 - ▶ Adverse impact on education
 - ▶ Adverse impact on health
 - ▶ Food issues
 - ▶ Impact on the environment
 - ▶ Frictions over international migration

Goals and Objectives: Toward a Consensus

- ▶ Despite the widely diverging and conflicting opinions, there is some common ground on the following:
 - ▶ Population is not the primary cause of lower living levels, but may be one factor
 - ▶ Population growth is more a consequence than a cause of underdevelopment
 - ▶ It's not numbers but quality of life
 - ▶ Market failures (as detailed on previous slides)
 - ▶ Voluntary decreases in fertility is generally desirable for most developing countries with still-expanding populations

Goals and Objectives: Toward a Consensus (Continued)

- ▶ Some Broader Policy Approaches
 - ▶ Attend to underlying socioeconomic conditions that impact development
 - ▶ Family planning programs should provide education and technological means to better regulate fertility
 - ▶ Consider policy in light of market failures
 - ▶ Tailor policy design to underlying (broadly speaking, economic, or socio-economic) behavior
 - ▶ Developed countries have responsibilities too

Policy Responses to High Fertility: Lessons from Household Models

- ▶ Fertility may be lowered with:
 - ▶ Improved women's education, role, and status
 - ▶ Female nonagricultural wage employment
 - ▶ Rise in family income levels through shared growth
 - ▶ Reduction in infant mortality, better health care
 - ▶ Development of old-age and social security plans
 - ▶ Expanded schooling opportunities, lowered real costs
 - ▶ Lowered prices and better information on contraceptives
 - ▶ Direct incentives such as subsidy benefits
 - ▶ Policies that have the effect of reducing boy preference
- ▶ The above list provides a framework for policy

Some Broader Policy Approaches

- ▶ What Else Developing Countries Can Do
 - ▶ Persuasion through education
 - ▶ Family planning programs
 - ▶ Address incentives and disincentives for having children through the principal variables influencing the demand for children
 - ▶ Coercion is not a good option
 - ▶ Raise the socioeconomic status of women
 - ▶ Increase employment opportunities for women (increases opportunity cost of having more children, as in microeconomic household theory)

Policy for Still-Developing Countries Facing Population Declines

- ▶ Over half of world population now in countries with below-replacement fertility
- ▶ The list includes China and a growing number of other developing countries
- ▶ Populations could decline before reaching high income levels - known as the problem of “getting old before getting rich”
- ▶ High dependency in reverse: large fraction retired, but incomes of the working-age insufficient to support them, or leave enough savings to invest sufficiently
- ▶ Five interrelated policy responses have been proposed
- ▶ Immigration with a well-regulated guest worker system and path to citizenship
- ▶ Retraining: Provide skills for working productively later in life
- ▶ Accelerate productivity growth to support a larger nonworking population
- ▶ Create conditions to attract foreign investment of types that can lead to increased worker productivity and incomes
- ▶ As a last resort, pro-natal incentives for families to have more children but such policies introduce distortions, are expensive, and may be unsustainable

Some Broader Policy Approaches (Continued)

- ▶ What the Developed Countries Can Do Generally
 - ▶ Address resources use inequities
 - ▶ More open migration policies
- ▶ How Developed Countries Can Help Developing Countries with Their Population Programs
 - ▶ Research into technology of fertility control
 - ▶ Financial assistance for family planning programs

Chapter Summary

- ▶ World population grew slowly for millennia and then surged after 1800; growth is now concentrated in developing regions, especially Africa, and population momentum keeps it going even as fertility falls
- ▶ The demographic transition moves countries from high birth and death rates (Stage I) through rapid growth (Stage II) to low, stable rates (Stage III)
- ▶ The Malthusian population trap ignores technological progress and household choice, but explains why we still worry about fertility in poor, fragile settings
- ▶ The microeconomic household theory treats children as consumer and investment goods: women's education and employment, lower child mortality, schooling and old-age security lower the demand for children
- ▶ Policy consensus: population is not the main cause of underdevelopment, but voluntary fertility decline is desirable; effective policy works through incentives, not coercion, and developed countries have responsibilities too

Concepts for Review

- ▶ Birth rate
- ▶ Death rate
- ▶ Demographic dividend
- ▶ Demographic transition
- ▶ Doubling time
- ▶ Family-planning programs
- ▶ Fertility rate
- ▶ Hidden momentum of population growth
- ▶ Life expectancy at birth
- ▶ Malthusian population trap
- ▶ Microeconomic theory of fertility
- ▶ Mortality rate
- ▶ Natural increase
- ▶ Net international migration
- ▶ Population-poverty cycle
- ▶ Population pyramid
- ▶ Rate of population increase
- ▶ Reproductive choice
- ▶ Total fertility rate (TFR)
- ▶ Under-5 mortality rate
- ▶ Youth dependency ratio